A brief review of software verification for quantum programs

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Abstract

Quantum program software verification is an important but challenging task. This article briefly reviews formal proofs methods and simulation methods for quantum program software verification. Formal proofs methods mainly have three categories and simulation methods have two main research directions.

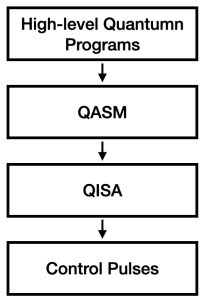
1 Introduction

As new quantum algorithms and new quantum computing devices emerge [1–4], more reliable and scaleble quantum computation systems are needed. Besides the theoretical challenge which requires a quantum algorithm to be at least polynomial faster than the classical counterpart [5], quantum algorithm designers also have to face the fact that there's no panacea for developing and verifying their algorithms, not only because different quantum computing devices have different controlling methods, but also the real devices that can run the algorithm may haven't emerged [6, 7].

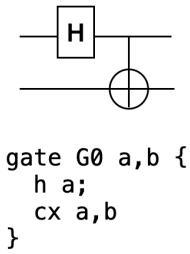
Quantum programming languages(QPLs)[1] are designed to tackle these challenges. In a broad sense, QPLs express quantum algorithms and how they are lowered to instructions that can be exactly executed on a real quantum computing device [9]. Currently, QPLs mainly have three levels, as shown in figure 1a. Quantum Programs can describe quantum algorithms [10–12], some programs can also describe the control of the host CPU [13] or the device constraints [14]. QASM is a gate-level desciption language that specifies 1- and 2-qubit operations [15, 16]. Then the QASM will be translated to QISA which basically contains instructions that control the pulses executated on quantum computing devices.

Writing a correct quantum program is an essential but challenging task. More and more quantum programming languages are being developed to deal with this challenge. A key question is: how to verify the program? This task is termed *software verification*. Software verification has two main purposes: (i) high-level quantum programs are correct syntactically and semantically [17–22] (ii) transformation on QPLs preserve equivalence [23–25]. Mainly three methods are used to achieve these two goals: (i) proofs via formal verification (ii) tracing via classical simulation (iii) quantum property testing. We will review the first two methods, and invite readers interested in the third method to refer to[26], because the first two methods are more closely related with the software verification.

[1] Use QPLs to distinguish from QPL in [8]



(a) QPLs stack [27]



(b) QASM example [15]

Figure 1: QPLs explanation

2 Formal proofs methods

Formal proofs express a quantum program in a well-defined semantics and use classical formal verification tools to efficiently verify the program. It's an open problem how to formalize the quantum circuit characteristics, such as the fragile qubits, no-cloning property, measurement, equivalent transformation of the gates (an example in figure2) into classical logic [5]. Many tools have been developed to do formal proofs on quantum circuits. I will categorize them based on the classical logic framework they are reduced to, because such final logic will express how the essential quantum concepts in circuits are modeled [28]. This section is organized as follow: each category's motivation is introduced first and then the core idea is explained. In order to emphasize the insights behind them, detailed explanation of syntax and semantics is omitted. To the best of the author's knowledge, no article has classified formal proofs methods from this view.

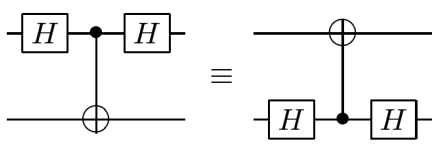


Figure 2: Equivalent transformation example

2.1 SMT

2.1.1 Motivation

Satisfiability Modulo Theories (SMT) refers to the problem of determining whether a first-order formula is satisfiable with respect to some logical theory[29]. For example, an SMT solver may want to check if

$$(x + y < 0) \wedge (x < 0) \wedge (y < 0) \tag{1}$$

is satisfiable. Different from Boolean satisfiability problem (SAT), SMT requires *background theory*. Different from general-purpose reasoning methods that want to find whether there exists a nonstandard interpretation of the symbols $+, <, 0$ that makes 1 satisfiable, practical SMT solvers just need to verify 1 under the assumption that $<$ is the usual ordering over the real number, $+$ the usual addition and 0 the identity of addition.

Such nature of SMT fits the QPLs verification well in the models of [23, 24]. In general, they use denotational semantics to model the quantum circuit and formulate the equivalent transformations as the background theory, then a circuit and the compiler passes exerted on them can be written to a first-order formula. Such formula can be solved efficiently using state-of-art SMT solvers, such as Z3[30].

2.1.2 Core idea

[23, 24] both intend to detect bugs of compiler passes in [31]. Such task has three main challenges. (i) Quantum compilers have to obey quantum mechanics rules which are complicated and counter-intuitive (ii) Quantum compilers rely heavily on the optimization passes to fulfill the device and noise budgets[32] (iii) Verification methods have to be easily scaled and automated, because the compiler code base keeps growing[31].

For example, in CertiQ[23] a verification can be modeled as below. For a compiler pass $app(C; CX; CX; T(C^{'}), Q)$. The claim G is $app(C; C^{'}, Q) \equiv app(C; CX; CX; T(C^{'}), Q)$. Two adjacent CX gates will cancel, so the first background theory P_1 is $\forall Q_1, app(CX; CX, Q) \equiv Q_1$. T is an equivalent transformation from the CertiQ library, so the second background theory P_2 is $\forall C_2\ Q_2, app(T(C_2), Q_2) \equiv app(C_2, Q_2)$. CertiQ can encode them into an SMT formula and feed it to Z3 to check $P_1 \wedge P_2 \wedge \neg Q$. CertiQ also designs an equivalence checker between different data representations for qubits, and a whole code stack to support both verification and execution at as low user burden as possible. Because SMT doesn't support unbounded loop structure, CertiQ also distills three loop patterns from most of Qiskit compiler passes. Giallar[24] builds upon CertiQ to enable a push-button style verification of the compiler passes.

2.2 Linear Type Systems

2.2.1 Motivation

Linear logic[33] restricts times premises are used to derive conclusions. To be specific, every premise has to be used exactly **once** to construct the conclusion. For example, if we have three premises $A \rightarrow B, A \rightarrow C$, and A in ordinary logic, it's safe to derive $A \mapsto B \wedge C$, even though A is used twice to derive the final logic. However, in linear logic the final logic is wrong because A is used twice. Instead, it allows $A, A \mapsto B \bigotimes C$ and $B \bigotimes C$ means it has both B and C. This article[2] uses a "chemical reaction matter-conserving" metaphor to describe linear logic. Such logic, in its nature, extremely resembles the quantum circuit where a qubit can only be used **once** if we don't consider the compilcated and error-prone qubit resue process.

2.2.2 Core idea

[20, 21] both use linear type system to assist verification proofs. In Qwire[21] wires are treated linearly. That means a wire can be only used once. The following Qwire code describes the qubit cloning and is wrong because its equivalent logic $W \mapsto W, W$ violates the "matter-conserving" nature.

```
Definition clone W : Box W (W * W).
    box_ w => output (w,w)
```

Compared with CertiQ which only uses Coq[34] to prove the rewriting roles, Qwire is implemented natively in Coq. Each wire is described by a wire type and gates are indexed by a pair

[2] https://www.cs.cornell.edu/courses/cs6110/2017sp/lectures/lec30.pdf

of wire types. A wire type is constructed with patterns which consist of basic WTypes. Coq's dependent types can be used to constrain how patterns can be constructed and thus only allow legal wire types. To verify general quantum programs VOQC[20] designs SQIR and extends the Qwire. The key difference between Qwire and SQIR is how they define qubits: SQIR uses concrete indices into a global register while Qwire uses abstract type variables. Concrete indices make it easier to reason about qubits because they are naturally mapped to columns and rows in a density matrix. However, they bring about much inconvenience when compositing closed circuits. Quipper[13], Proto-Quipper-S[35], Proto-Quipper-M[17] and Proto-Quipper-D[36] are also based on a linear type system with extra QPLs related concepts that help verification. For example, Proto-Quipper-M starts by defining parameters and state, and defines categorical structures in the model. Such design philosophy ensures the language is "correct by design".

2.3 Floyd-Hoare logic

2.3.1 Motivation

The goal of Hoare logic is to provide a compositional method for proving the validity of specific Hoare triples. The standard notation of a Hoare triple is $\{P\}c\{Q\}$, which means if command c begins execution in a state satisfying assertion P, and if c eventually terminates in some final state, then that final state will satisfy the assertion Q[37]. P is called precondition and Q is called postcondition. Quantum programs can fit into such general logic framework. Basically, [18] defines the correctness formula $\models_{par} \{P\}S\{Q\}, \models_{tot} \{P\}S\{Q\}$ of a quantum program S as the inequality between P and Q. The precondition and the postcondition are defined as $tr(P\rho)$ and $trQ[S](\rho)$.

2.3.2 Core idea

[18, 19, 25] all use the Hoare logic framework to describe the verification problem. QHL [18] models quantum programs as super-operators and a quantum predicate as an Hermitian operator. In other words, P and Q in the Hoare triple are two quantum predicates and c is a quantum program. Based on the denotational semantics of quantum programs, the notions of partial and total correctness can be defined, which can further defines the weakest and the weakest liberal preconditions. aQHL [19] develops a practical variant of QHL. It restricts P and Q to projection operators which simplifies the inference rules. Besides, it develops several rules of reasoning about robustness of quantum programs to prove approximate satisfaction of a postcondition.

3 Simulation Methods

Such methods do simulation of quantum programs on a classical computer. Based on what the simulation returns, they can be divided into two categories: strong and weak [26]. Strong simulation returns the probabilities of the output measurement outcomes, while weak simulation only returns a sample of the measurement outcomes.

Designing a simulation algorithm is challenging, because it must simulate a quantum program both with high accuracy and with low error rate. What's more, for fairly large-scale quantum programs, it is infeasible to directly record state vectors at every simulation time step because of the exponential memory consumption. Therefore, most researches choose to simulate programs with specific structures and figure out simulation algorithms that fit the structures. For example, theoretically, for stabilizer circuits (only consisted of gates generated from Clifford group), there exists efficient classical simulation [38, 39].

3.1 Classical big data techniques help

Naive density matrix simulation algorithm requires exponential time and space cost. For a depth n circuit with m-qubit(no entanglement), a unitary gate is a $2^m \times 2^m$ matrix. So if we use naive matrix multiplication to simulate the unitary gate, we need to do $O(n2^{2m})$ arithmetic operations and store a $O(m)$ state vector. However, some big data algorithm from classical computation, such as data compression and distributed computing can be applied to accelerate the simulation [40]. Besides, researchers has surprisingly some circuits where all qubits are entangled, but still can be simulated efficiently on a classical computer [41].

3.2 New notions open new opportunities

Tensor network method use a tensor network to represent a quantum circuit, and gates are modeled as tensor network contraction [26]. Therefore, the challenge is transformed to efficient tensor contraction algorithm which is a well researched area [42]. Combined with Feynman's sum-over-path approach, undirected graphical model is established to further optimize the tensor contraction algorithm [43]. Other new models, such as ZX-diagrams [44, 45] have also been developed to accelerate simulation.

4 Other works

Because this review aims to summarize some insights when designing verification methods for QPLs, formal form of theoretical foundations of the mentioned methods aren't included. For example, two fundamental theorems: Exact compilation theorem and Solovay-Kitaev theorem [46] that make quantum compilation possible aren't discussed in this review. Besides, improvements on methods in this review are been developed. For example, Proq [47] uses projection operators to design runtime assertions. [48] further extends Proto-Quipper-M to use the linear dependent type theory. What's more, researchers are developing new IRs for QPLs to optimize quantum program correctly[49, 50]. Interestingly, the name "Paulihedral" is composed of "Pauli" and "Polyhedral". And polyhedral model[51] is a type of classical compiler optimization. Quartz[50] also uses TASO[52], an optimization tool for computation graph in classical programs to optimize quantum programs. It's reasonable to expect more knowledge from classical computer science to improve QPLs.

5 Conclusion

This article briefly reviews two software verification methods: formal proofs methods and simulation methods. Formal proofs methods can be categorized into three types according to the classical logic framework they are reduced to: SMT, linear type systems and Floyd-Hoare logic. All have their own motivation to be used. Simulation methods have two takeaway messages: classical big data techniques help and new notions open new opportunities. Due to the restriction of author's knowledge and the space, QPLs for each verification method are not fully listed and advanced discussions on the verification theory are not included.

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